

August 2026

VISION-LANGUAGE-ACTION MODELS FOR TRANSPARENT PLASTIC WASTE SORTING

Méa Vittot

H00512699

MSc Robotics

Heriot-Watt University

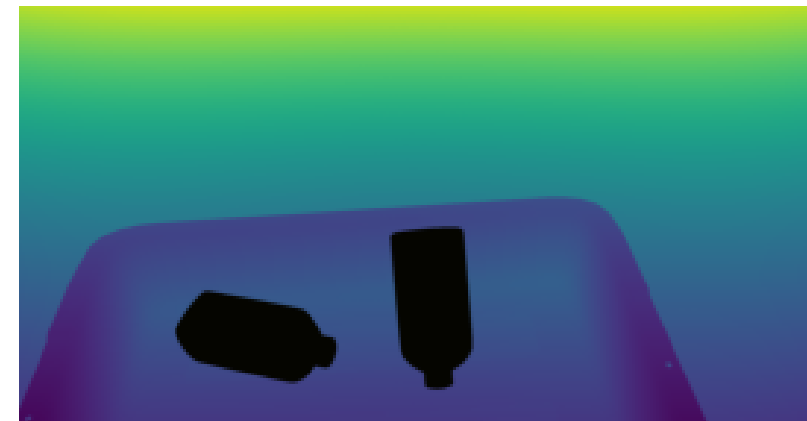
Supervisor: Dr. Maria Koskinopoulou



From roboticsbiz.com

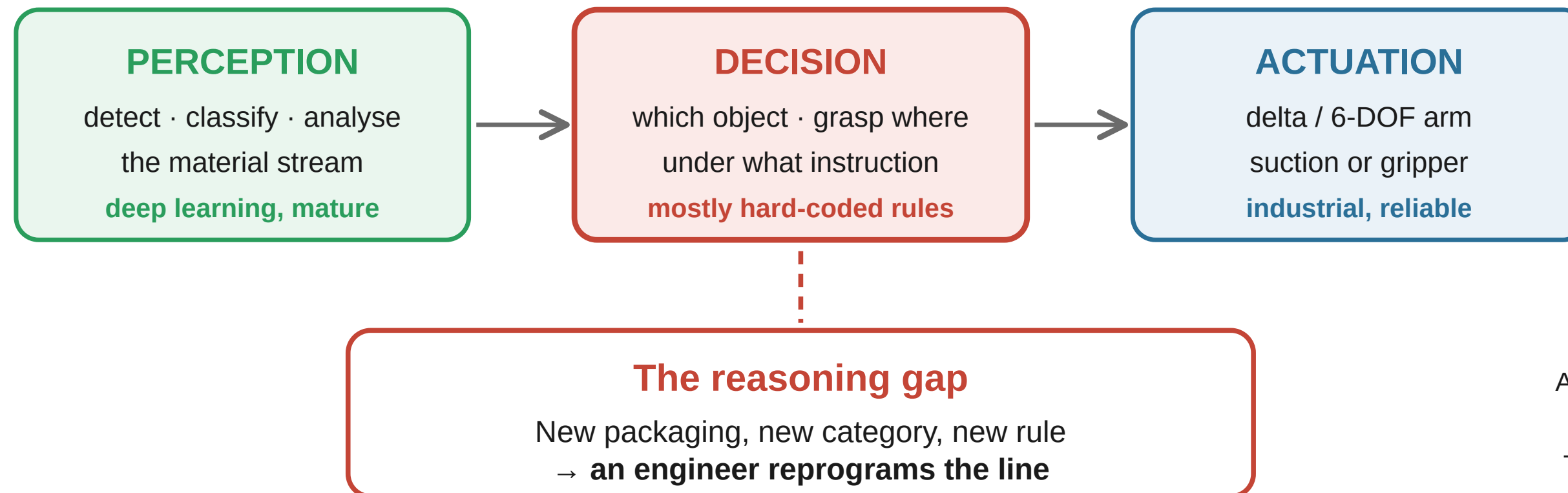
THE PROBLEM

- 1 Transparent PET plastic → depth sensors return nothing
- 2 **No dataset of this task exists**





INDUSTRIAL CONTEXT



Deployed today

AMP Robotics · Greyparrot
Recycleye · ZenRobotics
TOMRA · Waste Robotics

INDUSTRIAL CONTEXT



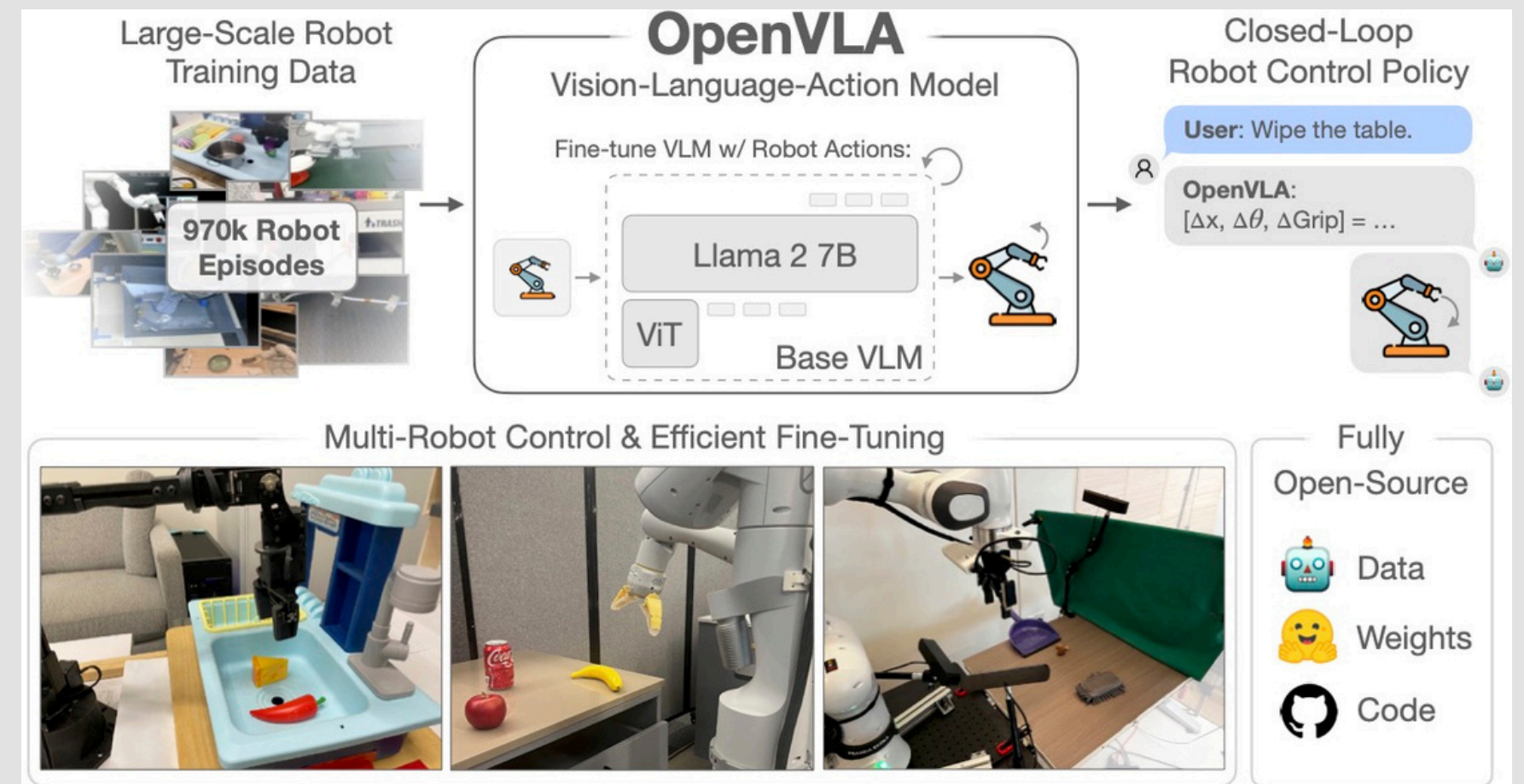
Recycleye Robot

WHY A VLA? WHY LANGUAGE?

Waste sorting needs semantic flexibility →

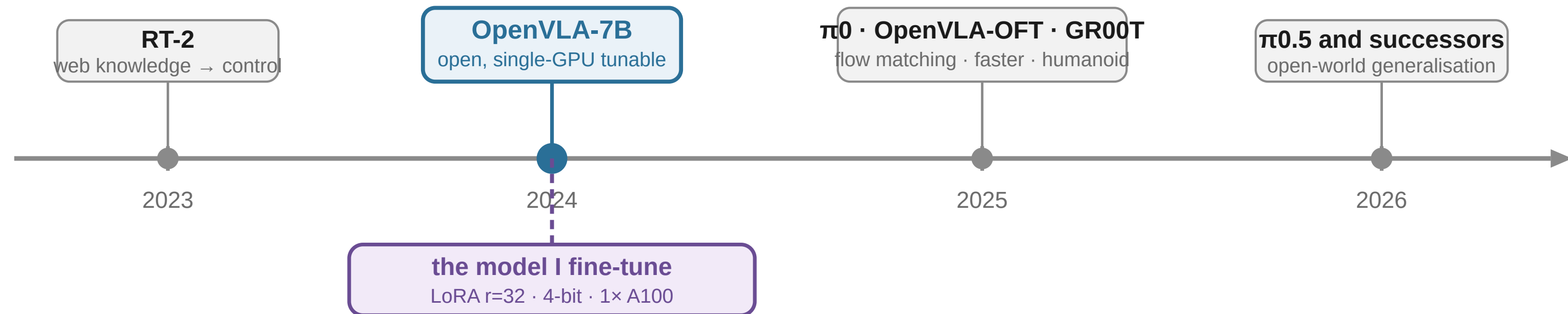


VLA = image + instruction => action •



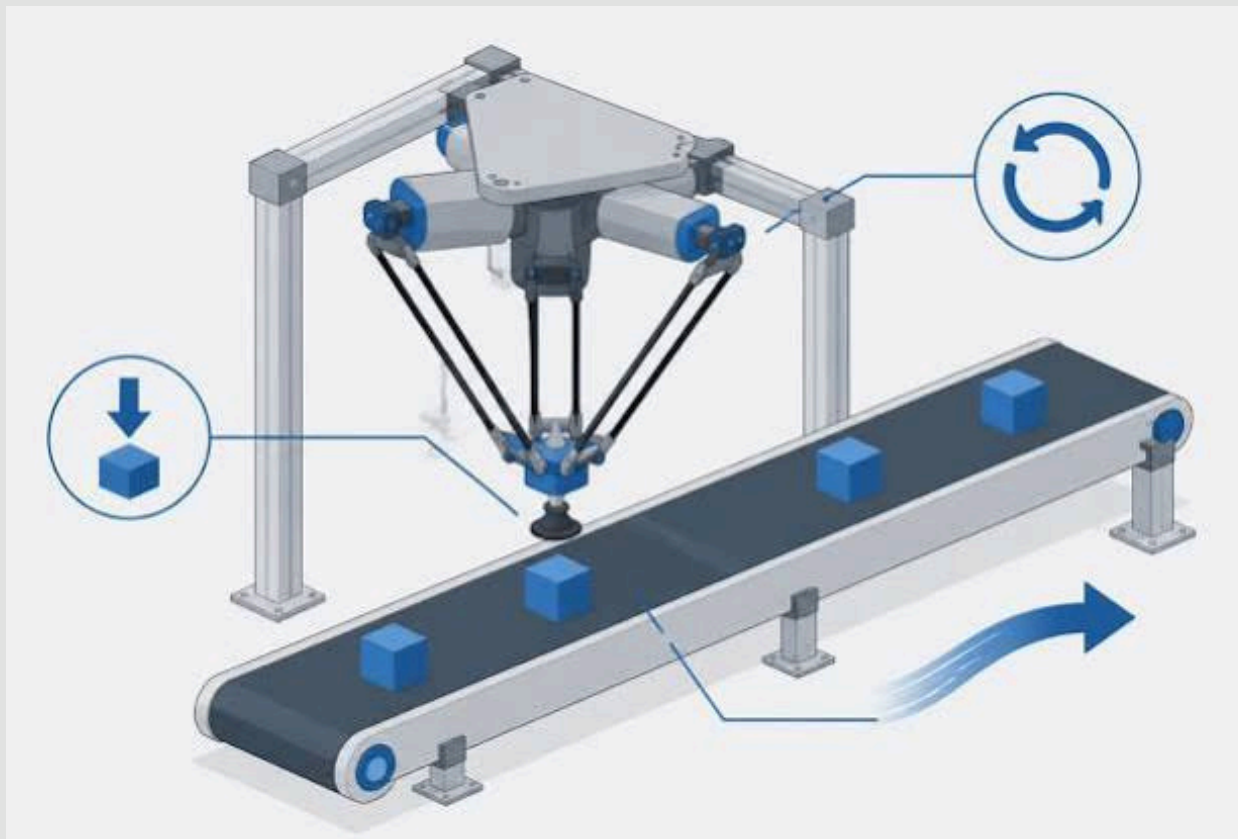
from openvla.github.io

STATE OF THE ART (VLA FRONTIER)



Strong semantics, weak spatial precision

SCOPE



1

Lab feasibility study — not industrial

2

Suction + Delta

=> top-down 2D point + orientation-free

- 5/7 DOF frozen

3

Transparent clean bottles from ClearGrasp.

A proxy, not real waste

OBJECTIVES & RESEARCH QUESTIONS

1

Feasibility: can composited **synthetic** data teach a VLA to **spatially** ground transparent objects (localise them well enough to grasp)?

2

Perception & depth : how well does RGB-only perception recover transparent geometry and generalise — and does normal-based depth reconstruction actually beat naive infilling?

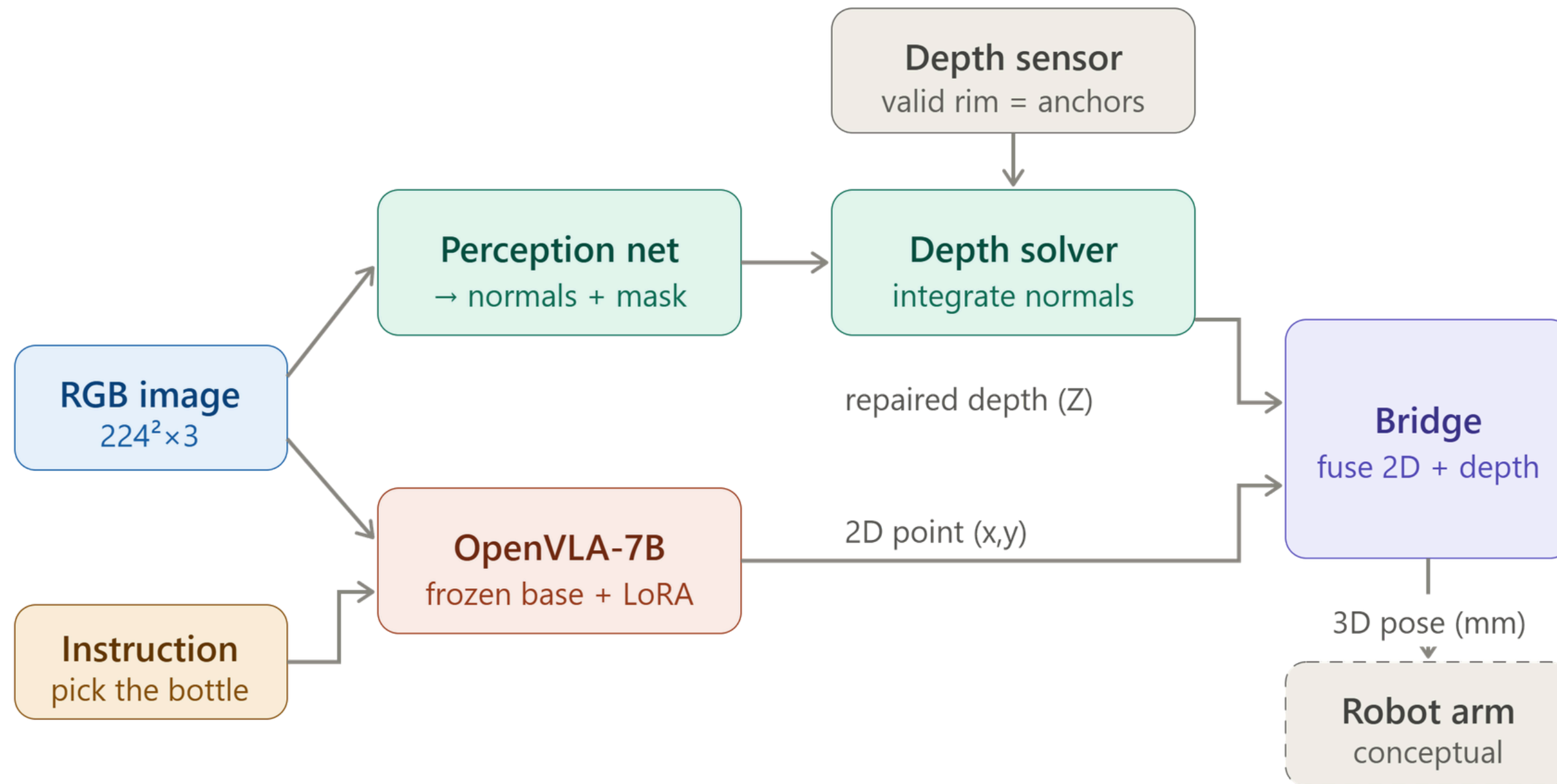
3

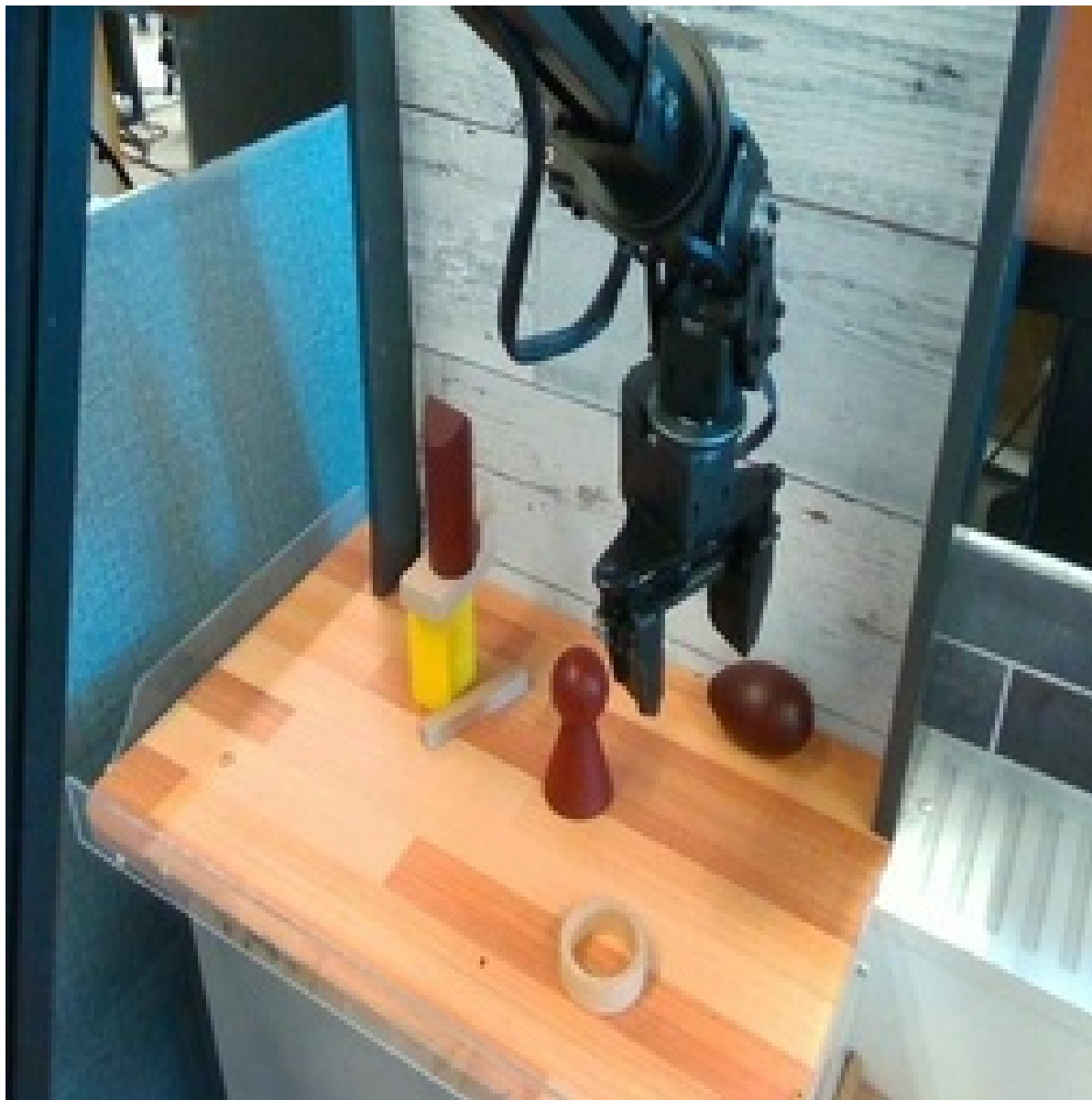
Semantics: does language/instruction conditioning survive fine-tuning — and if it collapses, can it be recovered within the trained instruction distribution?



METHODOLOGY

THE PIPELINE





WHY THE DATA HAS TO BE SYNTHETIC

VLA learns from demonstrations:
image + instruction + correct action

no data + no robot => synthetic



THE DATASET: CLEARGRASP + RESORT-IT

ClearGrasp gives: RGB · mask · normals · depth (11,702 renders)



ReSort-IT composites on 153 backgrounds + 50/50 native



- Compositing adds background variety
but deletes the refraction/shadow

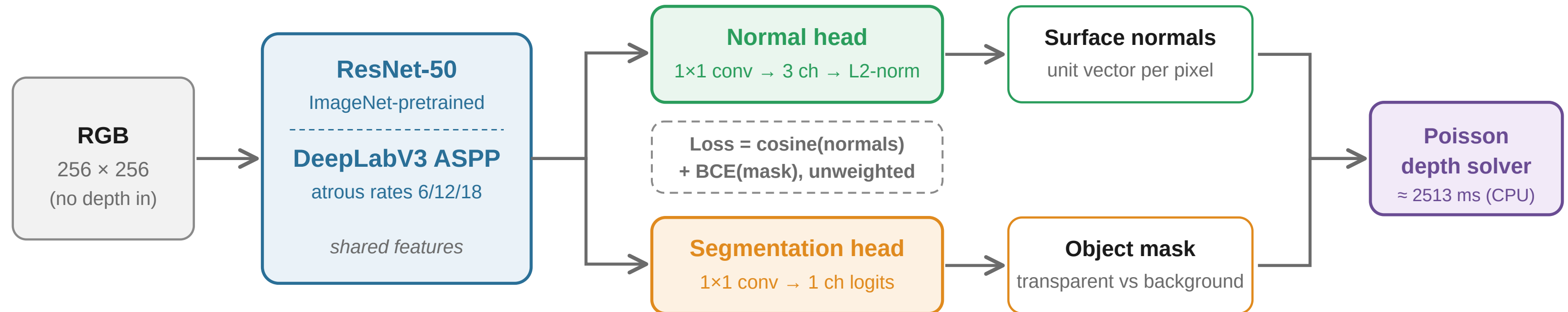
GRASP TARGET : GROUND TRUTH

Per-object centre + Validity check → 10% invalid centroids =>
fixed (point-in-polygon)



PERCEPTION MODULE

- Dual-head => normals + mask
- In-distribution (train-set fit): IoU 91.1% · normal error 19.6°



Training: AdamW · LR 1e-4 · batch 4 · 10 epochs · square-plastic-bottle category

DEPTH RECONSTRUCTION SOLVER

Poisson solver: integrate normals, anchored to rim depth => fills the hole

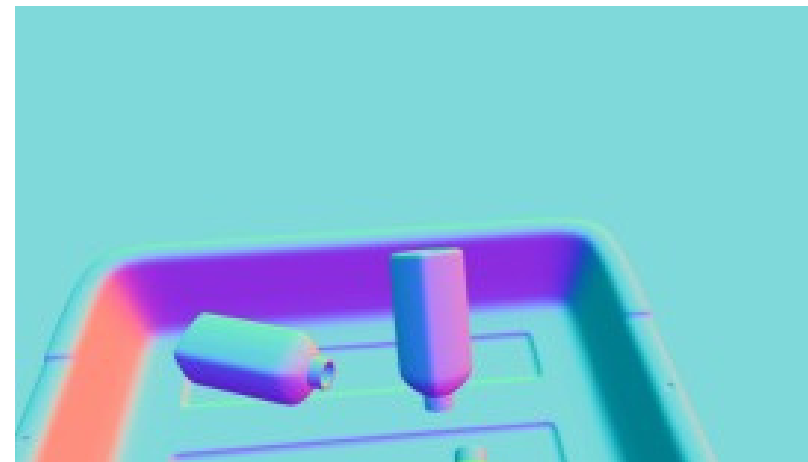
1 · RGB in



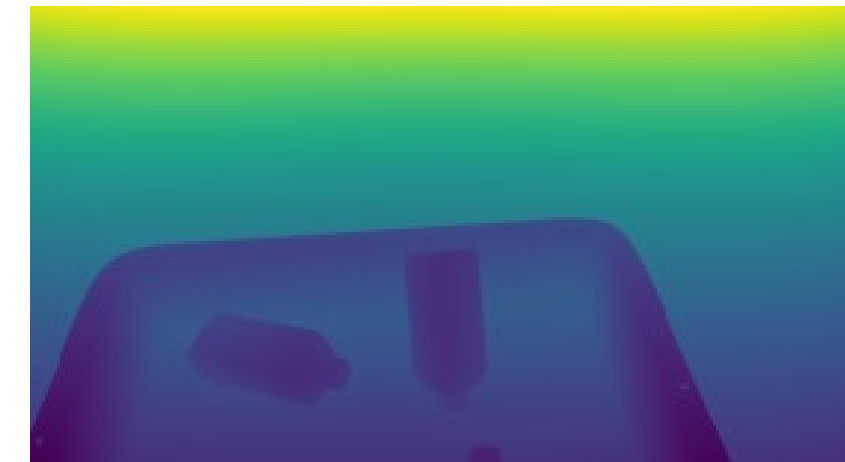
2 · Sensor depth
(hole over the object)



3 · Predicted surface
normals (the 'slopes')



4 · Reconstructed depth
(integrate slopes from the rim)



VLA FINE-TUNING + COMPUTE BUDGET

=> Used Conservative Training parameters

Frozen base — 7.56 B parameters (4-bit NF4)

LoRA trainable: 16.78 M = 0.22%

Rank32 · α 32 · dropout 0.05 · single A100 · ~26 h · **best val loss 0.163**

=> Also tried standard recipe

same validation loss, ~15× faster, spatial accuracy unchanged



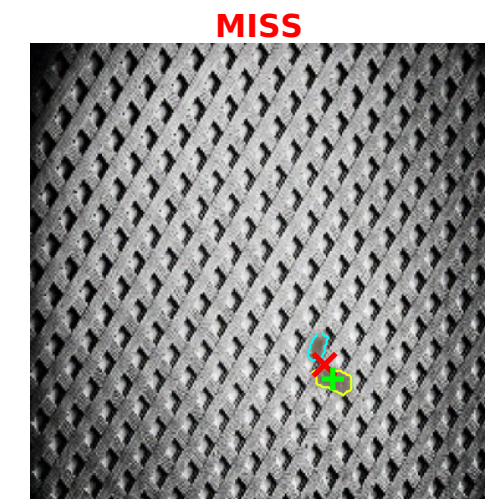
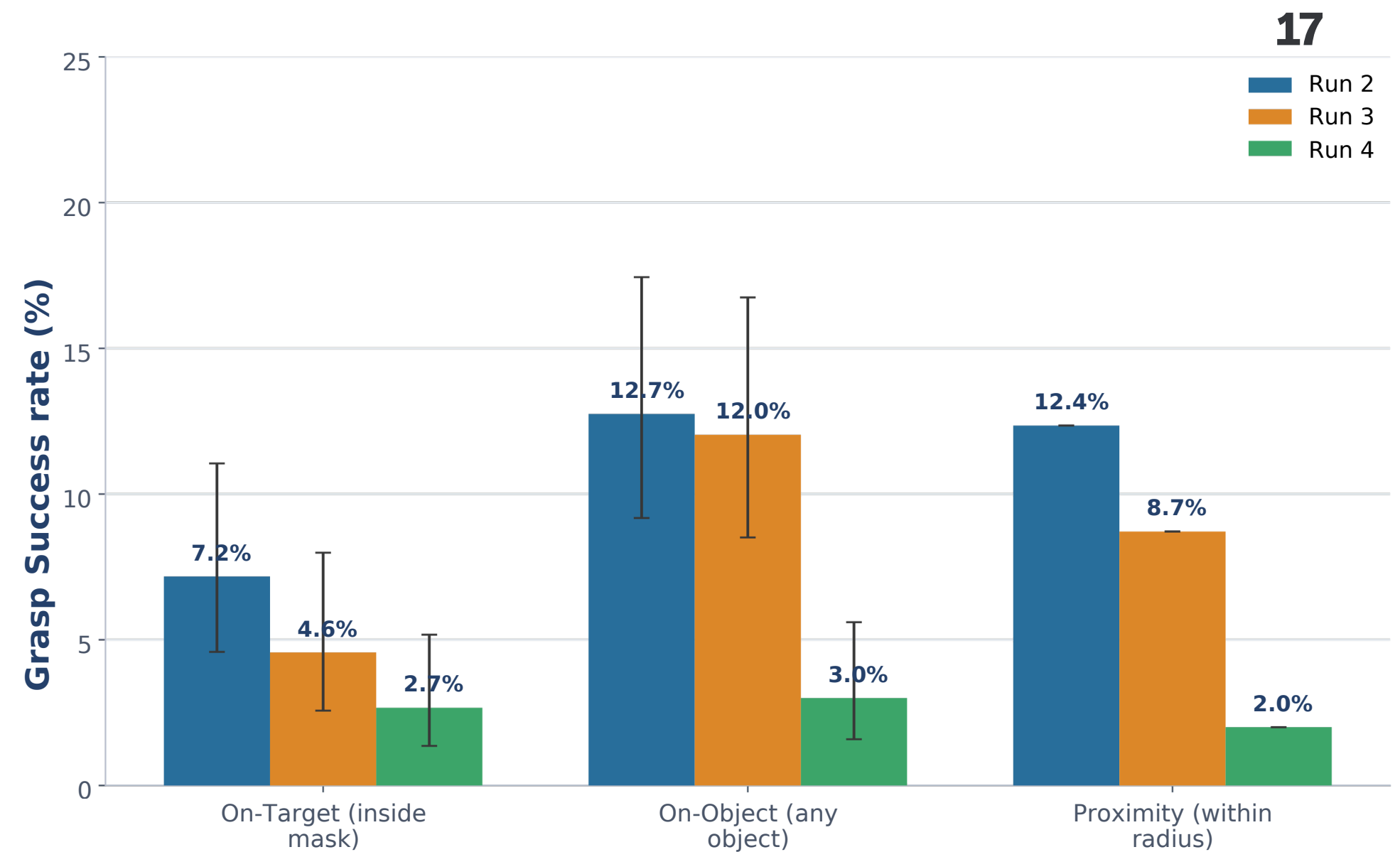
RESULTS

HOW THE SYSTEM IS EVALUATED

	Metrics	Data
GRASP targeting	ON-TARGET % (point-in-mask 2D proxy) => Policy outputs to a 256-bin grid (1 bin $\approx$ 0.875 px)	SYNTHETIC HELD-OUT
INSTRUCTION robustness	Parse-fail % (Strictly within ClearGrasp distribution)	SYNTHETIC HELD-OUT
DEPTH reconstruction	RMSE (m) against ground- truth depth	CLEARGRASP VAL

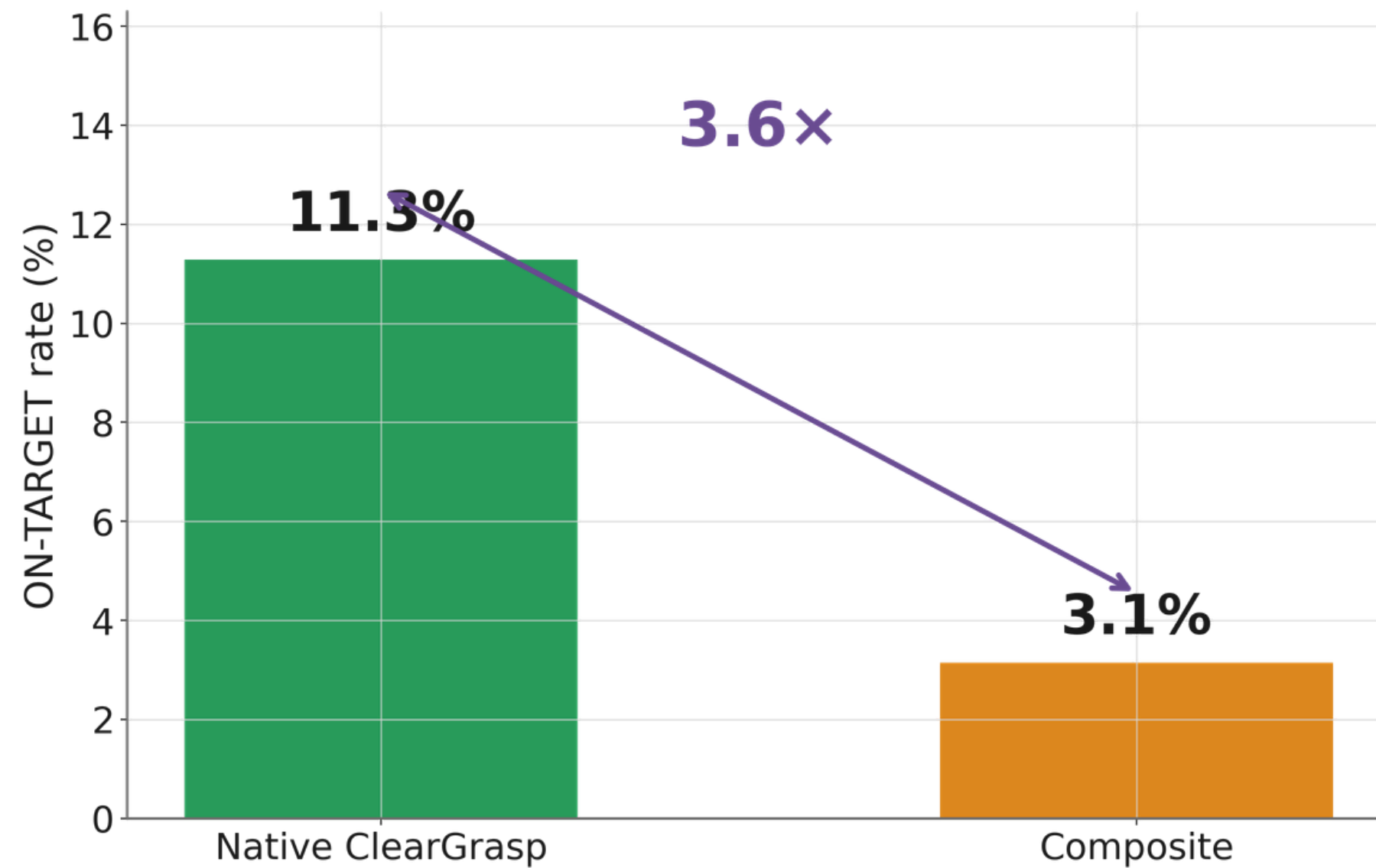
GRASP SUCCESS

- On-target 7.2% overall [3.9, 10.9]
- **Median 42 < Centre-prior 47 px** => real signal · but ~26% gross-miss tail



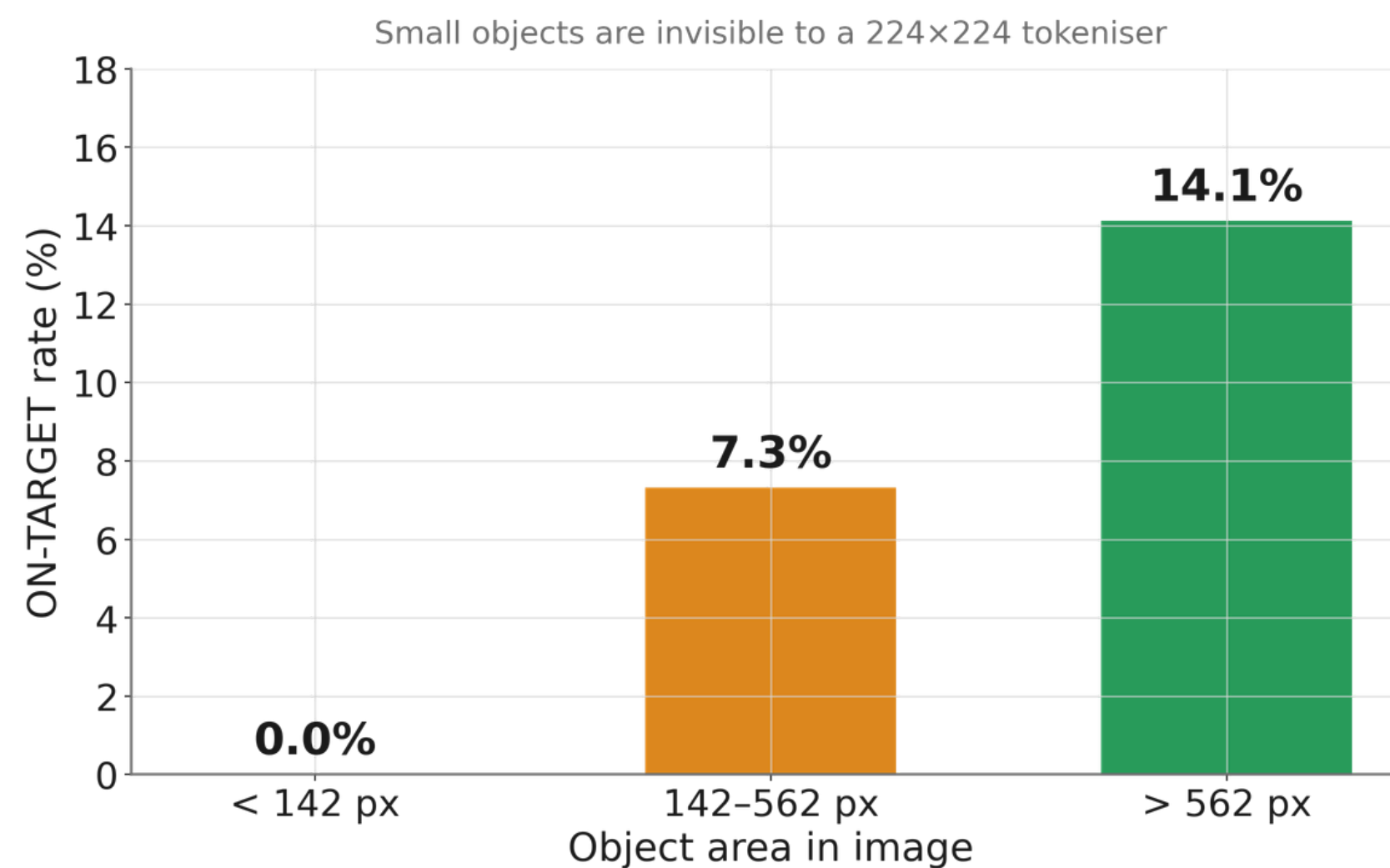
NATIVE VS COMPOSITE

Same model, same weights → composite data is the dominant problem

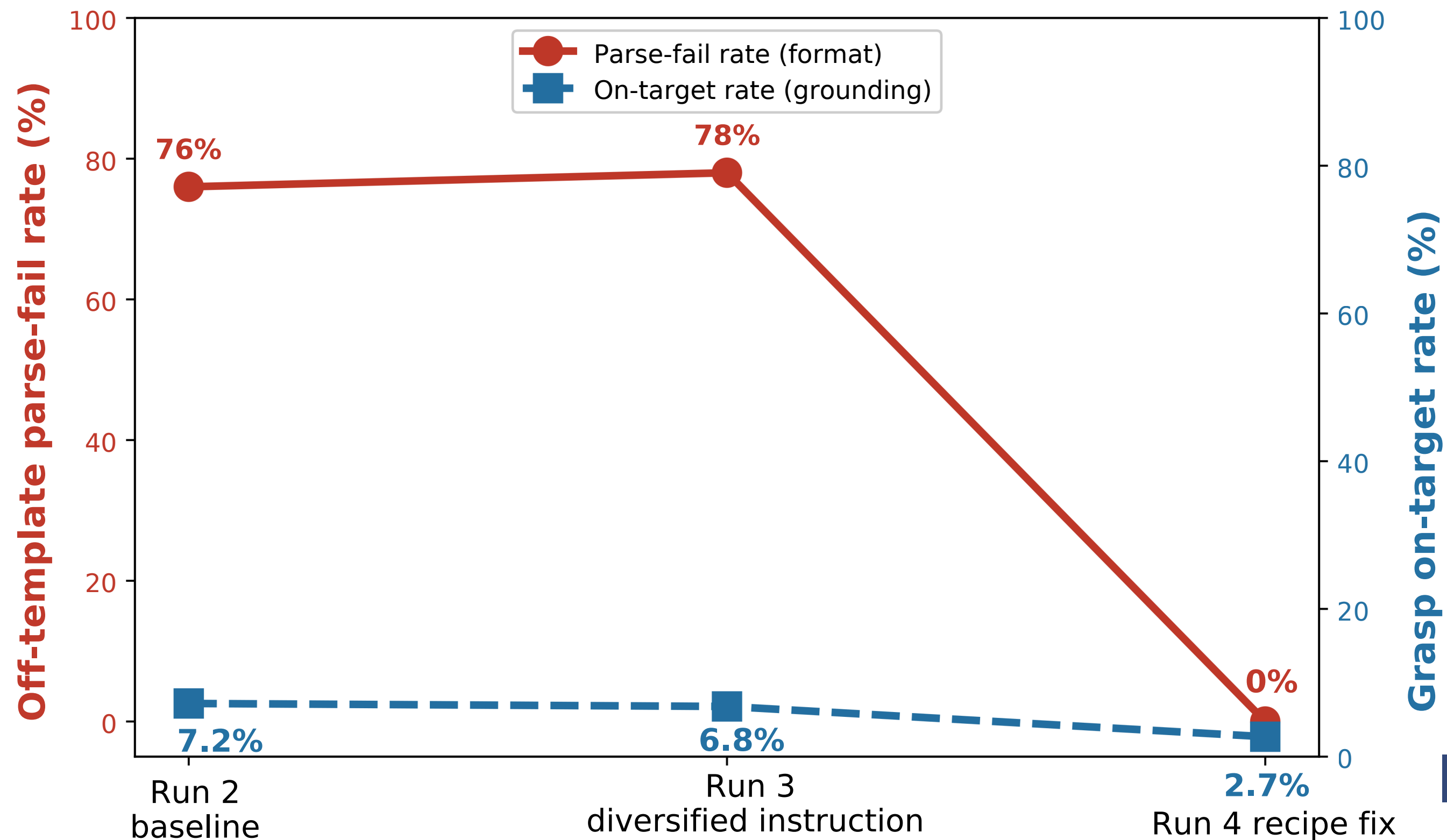


cluster CI [1.1, 15.9]

DATA SIZE TEST

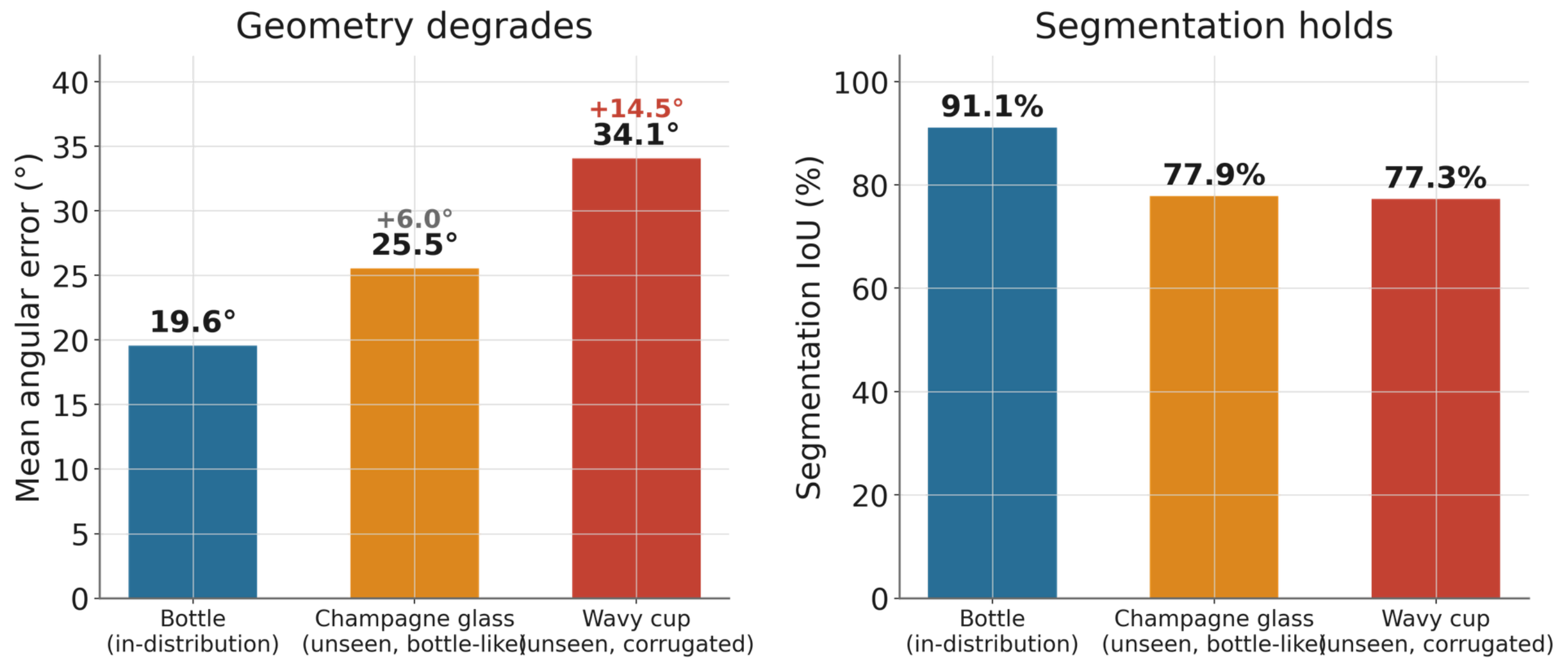


INSTRUCTION ROBUSTNESS

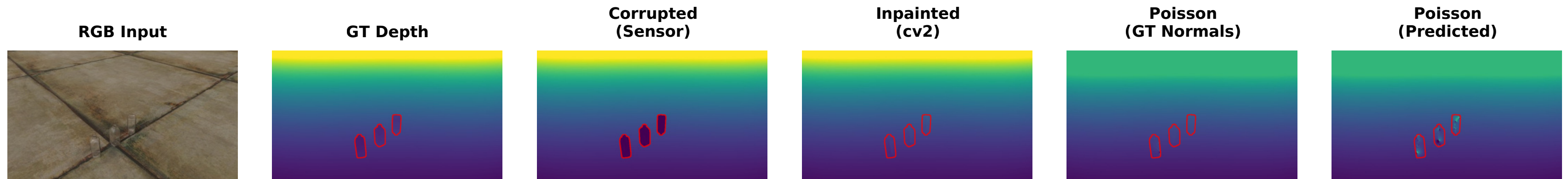


PERCEPTION RESULTS

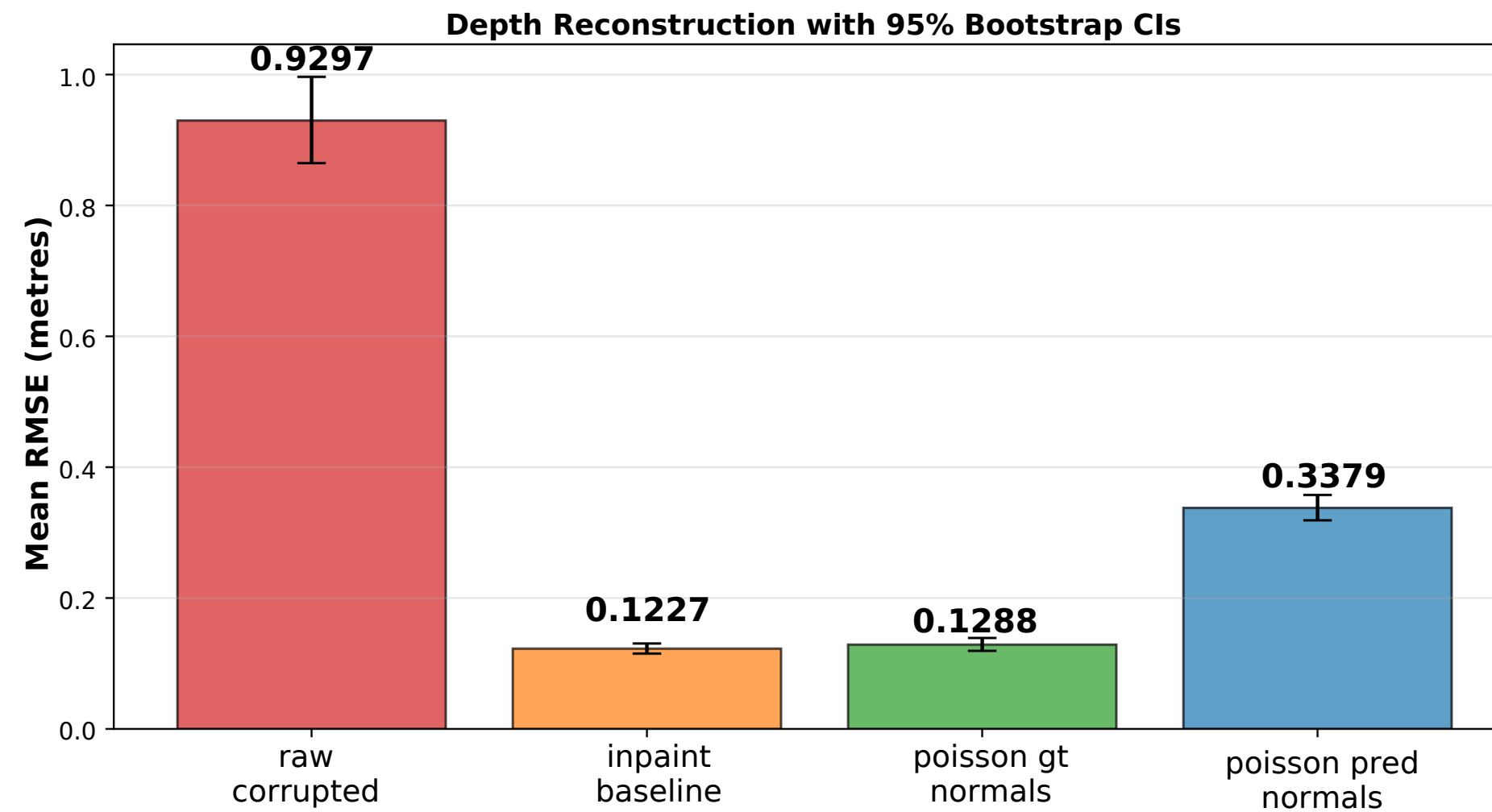
segmentation generalises (77% IoU on unseen shapes) • geometry degrades with shape difference from a bottle



DEPTH RECOVERY COMPARISON



With perfect normals, the solver ties with inpainting baseline ($p=0.64$). Real normal error costs 0.21 m



GENERALIZATION RESULTS

On ZeroWaste (real waste), the policy collapses:



- Outputs gibberish
 - 0% on-target
- (1) It learned a synthetic-specific pattern
(2) The Data domain gap is enormous :
ClearGrasp vs real waste

Perception transfers (~30% hit-precision)



SYNTHESIS

- The VLA learned to emit an action pattern
 - Bounded to clean ClearGrasp plastic
 - Every result = diagnostic for future work
 - The lever is data fidelity
-

FUTURE WORK

- Physically-based rendering
- Generalisation beyond the trained domain
- Fine-tuning recipe
- Depth-as-input to the VLA, not downstream
- Simulation validation: Gazebo (Unlock 6-DOF action dimensions)



**THANK
YOU**